Math 221 Programming Homework 2

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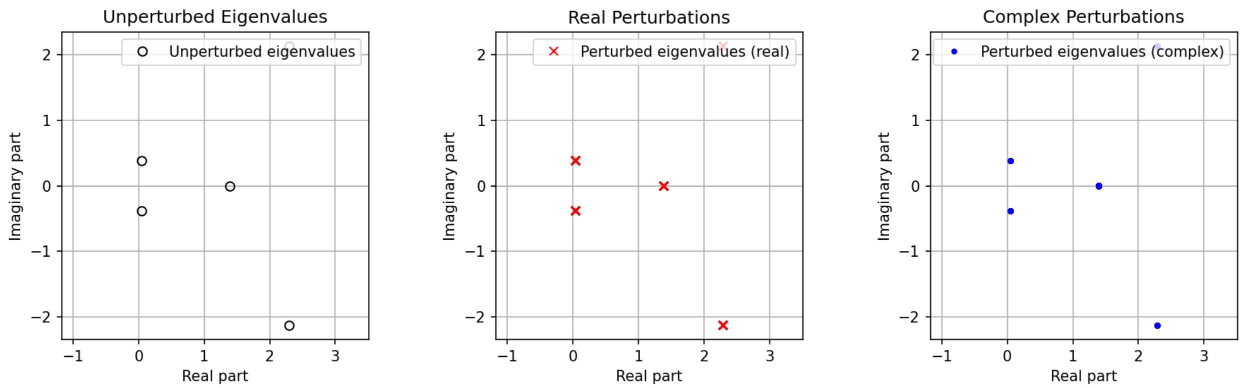
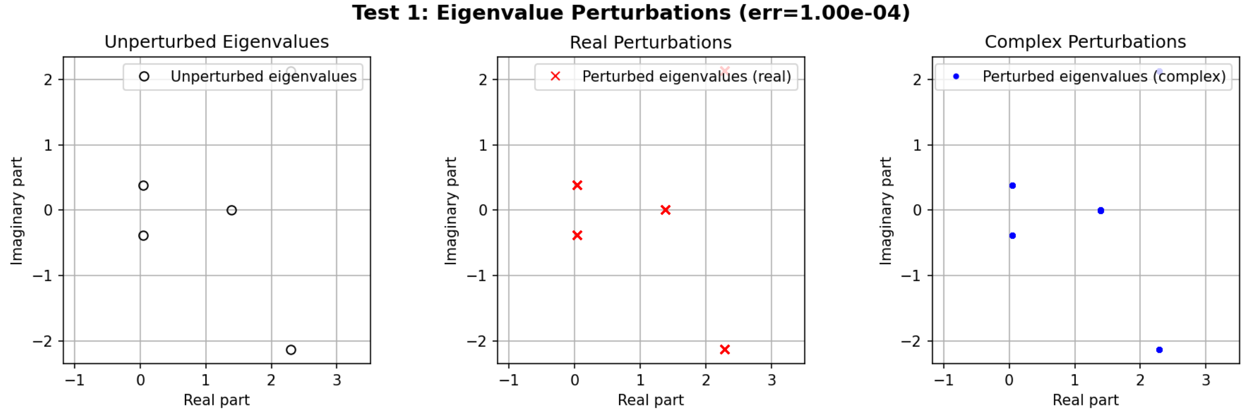
Fall 2025

Test 1: Eigenvalue Perturbations (err=1.00e-05)



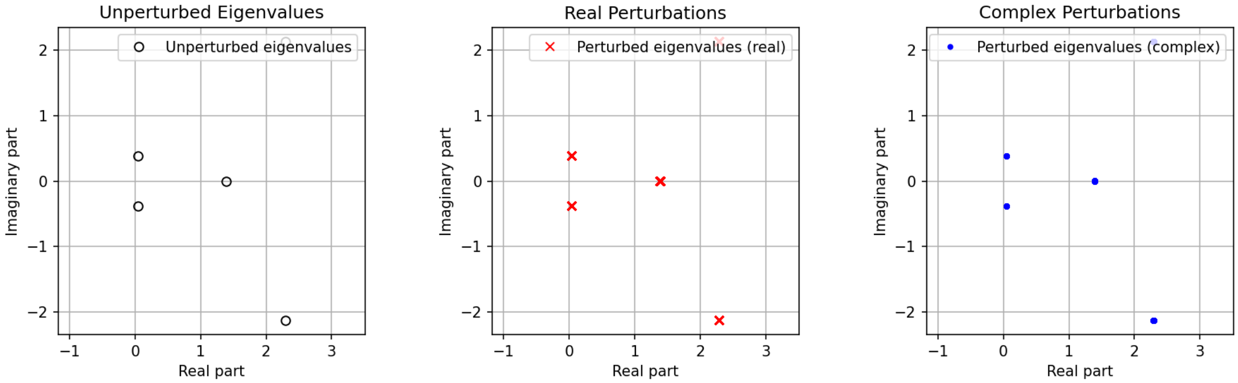
Eigenvalues	Condition Numbers
2.291307e+00+2.129147e+00j	1.341341e+00
2.291307e+00-2.129147e+00j	1.341341e+00
1.390963e+00+2.459512e-16j	2.009187e+00
4.221158e-02+3.844350e-01j	2.354910e+00
4.221158e-02-3.844350e-01j	2.354910e+00

Test 1: Eigenvalue Perturbations (err=1.00e-04)



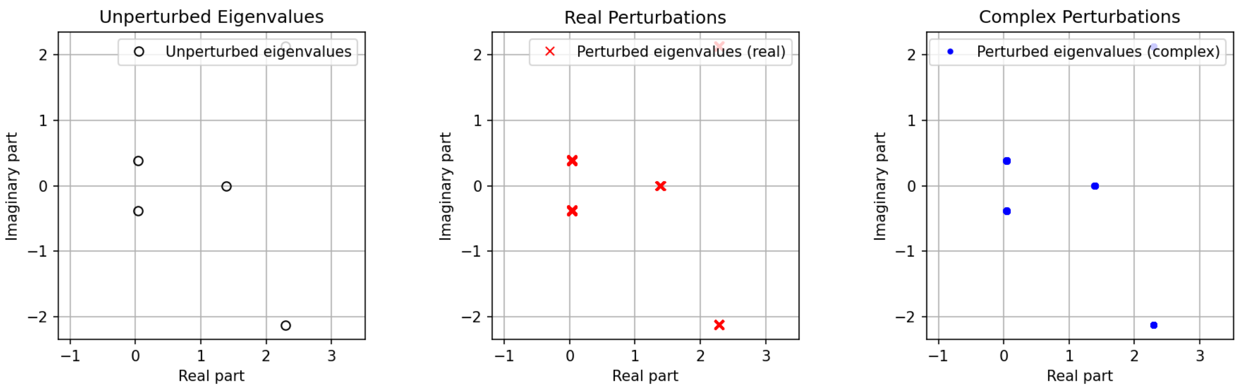
Eigenvalues	Condition Numbers
2.291307e+00+2.129147e+00j	1.341341e+00
2.291307e+00-2.129147e+00j	1.341341e+00
1.390963e+00+2.459512e-16j	2.009187e+00
4.221158e-02+3.844350e-01j	2.354910e+00
4.221158e-02-3.844350e-01j	2.354910e+00

Test 1: Eigenvalue Perturbations (err=1.00e-03)

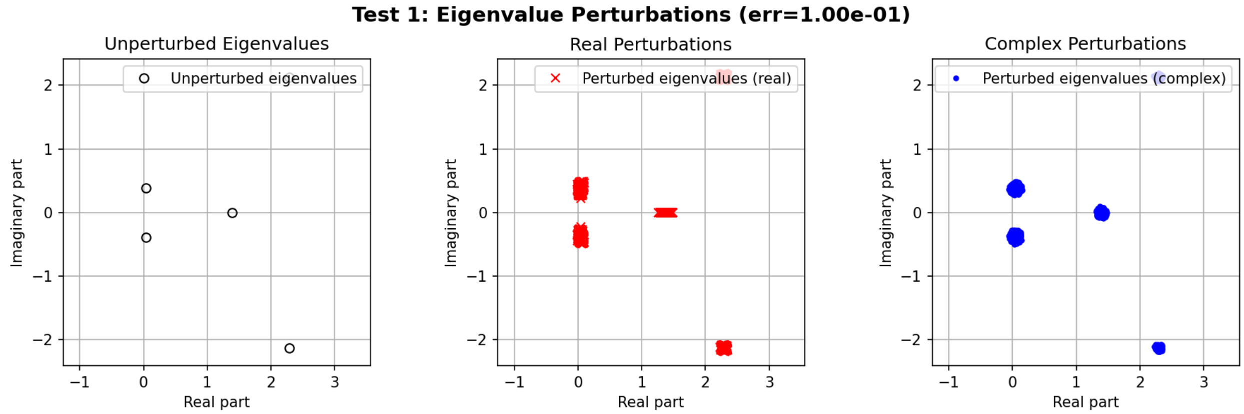


Eigenvalues	Condition Numbers
2.291307e+00+2.129147e+00j	1.341341e+00
2.291307e+00-2.129147e+00j	1.341341e+00
1.390963e+00+2.459512e-16j	2.009187e+00
4.221158e-02+3.844350e-01j	2.354910e+00
4.221158e-02-3.844350e-01j	2.354910e+00

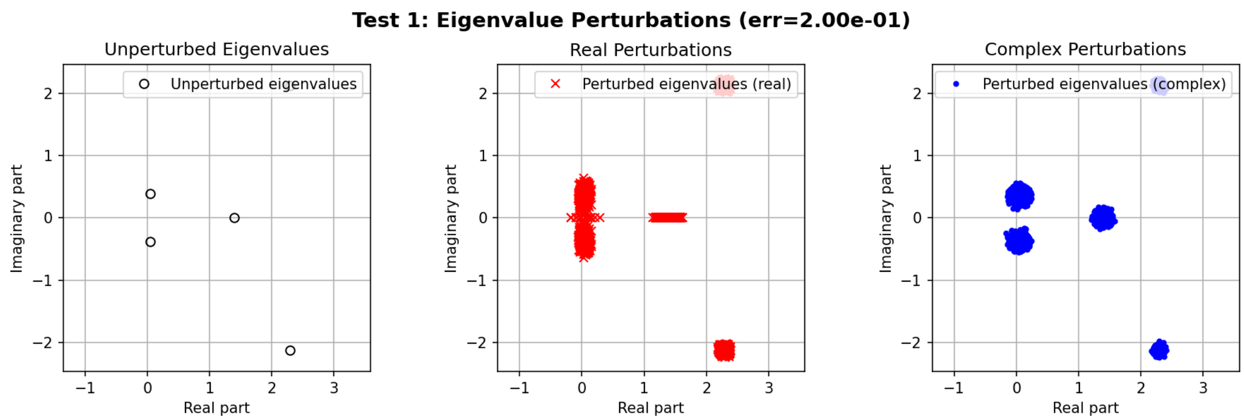
Test 1: Eigenvalue Perturbations (err=1.00e-02)



Eigenvalues	Condition Numbers
2.291307e+00+2.129147e+00j	1.341341e+00
2.291307e+00-2.129147e+00j	1.341341e+00
1.390963e+00+2.459512e-16j	2.009187e+00
4.221158e-02+3.844350e-01j	2.354910e+00
4.221158e-02-3.844350e-01j	2.354910e+00



Eigenvalues	Condition Numbers
2.291307e+00+2.129147e+00j	1.341341e+00
2.291307e+00-2.129147e+00j	1.341341e+00
1.390963e+00+2.459512e-16j	2.009187e+00
4.221158e-02+3.844350e-01j	2.354910e+00
4.221158e-02-3.844350e-01j	2.354910e+00



Eigenvalues	Condition Numbers
2.291307e+00+2.129147e+00j	1.341341e+00
2.291307e+00-2.129147e+00j	1.341341e+00
1.390963e+00+2.459512e-16j	2.009187e+00
4.221158e-02+3.844350e-01j	2.354910e+00
4.221158e-02-3.844350e-01j	2.354910e+00

1. **Comparison with the bounds from §4.3.** In Test 1 the unperturbed eigenvalues are

$$\lambda_1, \overline{\lambda_1} \approx 2.29 \pm 2.13i, \quad \lambda_2 \approx 1.40, \quad \lambda_3, \overline{\lambda_3} \approx 0.042 \pm 0.384i,$$

with eigenvalue condition numbers $\kappa_i = 1/|y_i^* x_i|$ between about 1.34 and 2.35 (from the tables under each plot). By the first–order perturbation bound and the Bauer–Fike theorem in §4.3, the eigenvalues of $A + \delta A$ must lie in discs centered at each λ_i of radius $|\delta\lambda_i| \lesssim \kappa_i \|\delta A\|_2 \approx \kappa_i\,\mathrm{err}$.:contentReference[oaicite:0]index=0 For Test 1 this gives radii of order $\kappa_i \mathrm{err} \approx (1.3$–$2.4)\mathrm{err}$. In the plots, for example at $\mathrm{err} = 2 \times 10^{-1}$, the complex–perturbation eigenvalues form roughly circular clouds around each unperturbed eigenvalue with diameter $\mathcal{O}(0.4$–$0.5)$, which is consistent with these predicted discs; as err is decreased to $10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}$, the radius of each cloud shrinks approximately in direct proportion to err, again in line with the linear bound $|\delta\lambda_i| \lesssim \kappa_i \mathrm{err}$.

2. **Real versus complex perturbations.** For *complex* perturbations, the random perturbation directions in $\mathbb{C}^{n\times n}$ are essentially isotropic, so the first–order formula $\delta\lambda_i \approx y_i^* \delta A x_i/(y_i^* x_i)$ produces perturbations in all directions in the complex plane. This is why the complex–perturbation plots look like small, nearly round blobs centered at each λ_i; they give a good picture of the full pseudospectrum for that radius. For *real* perturbations, we restrict to real δA, so complex eigenvalues occur in conjugate pairs. The perturbations of a conjugate pair are coupled, and only a subset of directions in the complex plane is sampled. In the plots this shows up as more anisotropic, often elongated clusters (for instance, the pair near $0.04 \pm 0.38i$ tends to move mostly in the imaginary direction under real perturbations), whereas the complex perturbations produce more symmetric, disc–like clouds around the same centers.

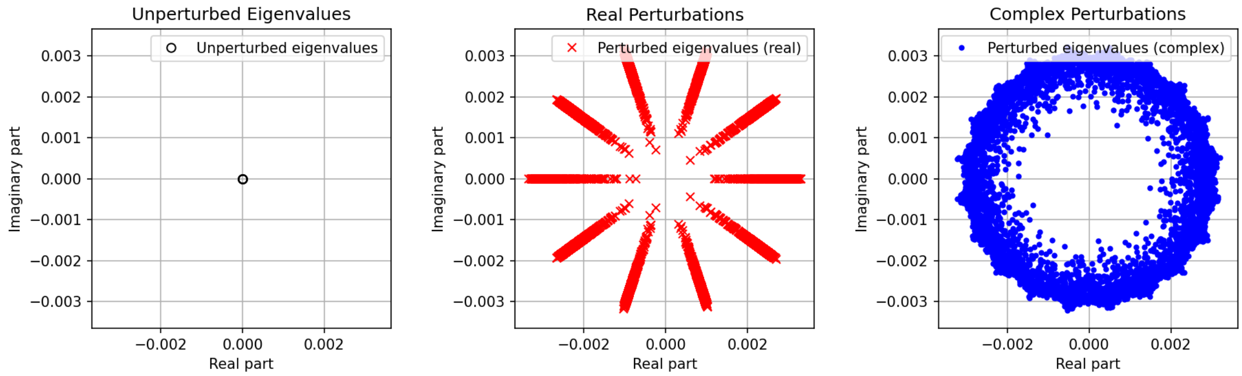
3. **Behavior as** $\mathrm{err} \to 0$**.** The successive Test 1 plots with $\mathrm{err} = 2\cdot 10^{-1}, 10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}$ show that the regions occupied by the eigenvalues shrink steadily toward the unperturbed eigenvalues. The diameter of each cluster scales essentially linearly with err, in agreement with the first–order theory $|\delta\lambda_i| \le \kappa_i \mathrm{err} + O(\mathrm{err}^2)$. In the limit $\mathrm{err} \to 0$ (ignoring roundoff), the regions would collapse to the exact eigenvalues. With floating–point arithmetic, the shrinking stops once err is comparable to the backward error of the eigenvalue algorithm, i.e. to a few units in the last place of the matrix entries.

4. **Limiting size and number of correct digits.** For a backward–stable eigenvalue algorithm applied in double precision with unit roundoff $u \approx 10^{-16}$, the effective perturbation size is $\|\delta A\|_2 \approx u\|A\|_2$. The perturbation bound then gives

$$|\delta\lambda_i| \lesssim \kappa_i \, u \, \|A\|_2.$$

In Test 1 the condition numbers are small, $\kappa_i \approx 1.3$–2.4, and $\|A\|_2$ is of order one, so the limiting cluster radius is $|\delta\lambda_i| \sim 10^{-16}$–$10^{-15}$. This corresponds to roughly $-\log_{10}(\kappa_i u) \approx 15$ correct digits in each eigenvalue. In other words, for Test 1 the eigenvalues are well–conditioned, and as $\mathrm{err} \to 0$ the pseudospectral regions shrink to machine–precision sized discs, so essentially all double–precision digits of the computed eigenvalues are reliable.

Test 2: Eigenvalue Perturbations (err=1.00e-12)




```
         Eigenvalues                  Condition Numbers
-------------------------------------------------------------
           0.000000e+00+0.000000e+00j      inf
           0.000000e+00+0.000000e+00j      inf
           0.000000e+00+0.000000e+00j      inf
           0.000000e+00+0.000000e+00j      inf
           0.000000e+00+0.000000e+00j      inf
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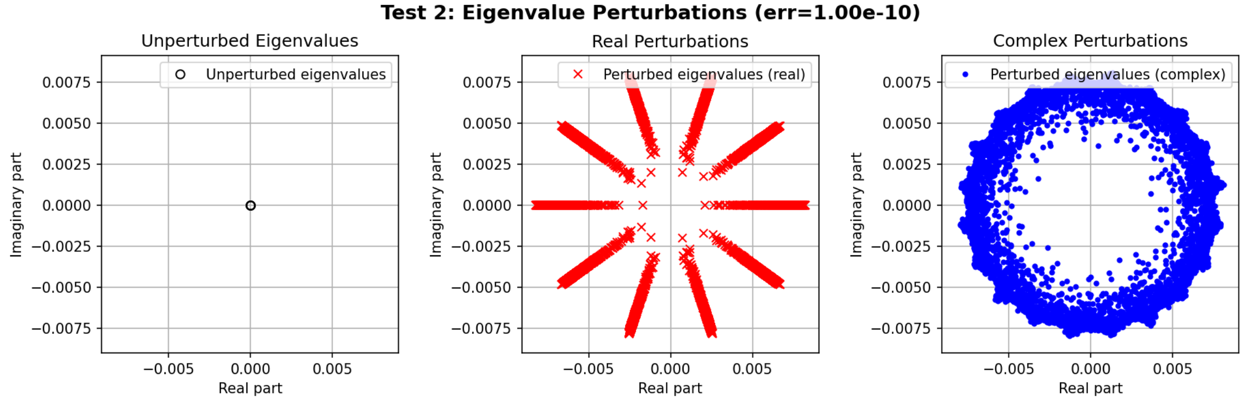
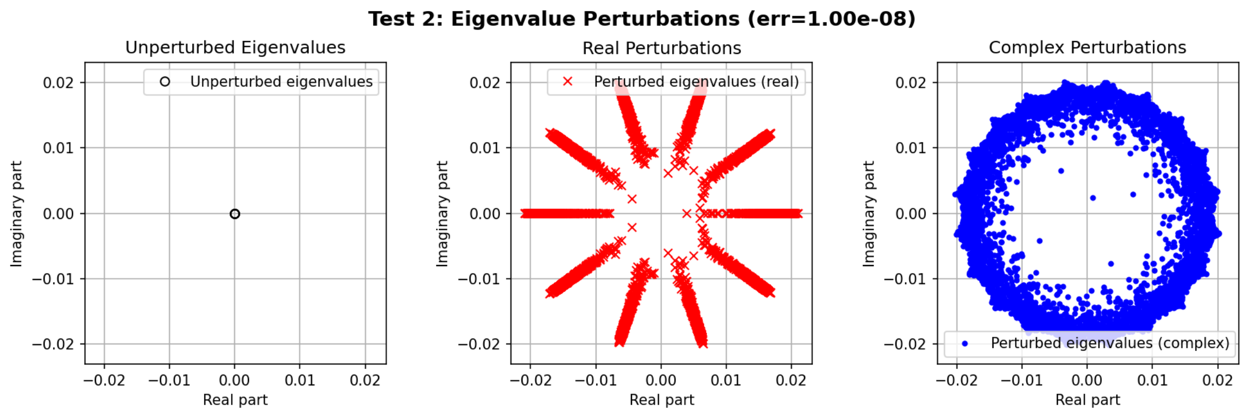
Test 2: Eigenvalue Perturbations (err=1.00e-10)




```
         Eigenvalues                  Condition Numbers
-------------------------------------------------------------
           0.000000e+00+0.000000e+00j      inf
           0.000000e+00+0.000000e+00j      inf
           0.000000e+00+0.000000e+00j      inf
           0.000000e+00+0.000000e+00j      inf
           0.000000e+00+0.000000e+00j      inf
```



Eigenvalues	Condition Numbers
0.000000e+00+0.000000e+00j	inf
0.000000e+00+0.000000e+00j	inf
0.000000e+00+0.000000e+00j	inf
0.000000e+00+0.000000e+00j	inf
0.000000e+00+0.000000e+00j	inf

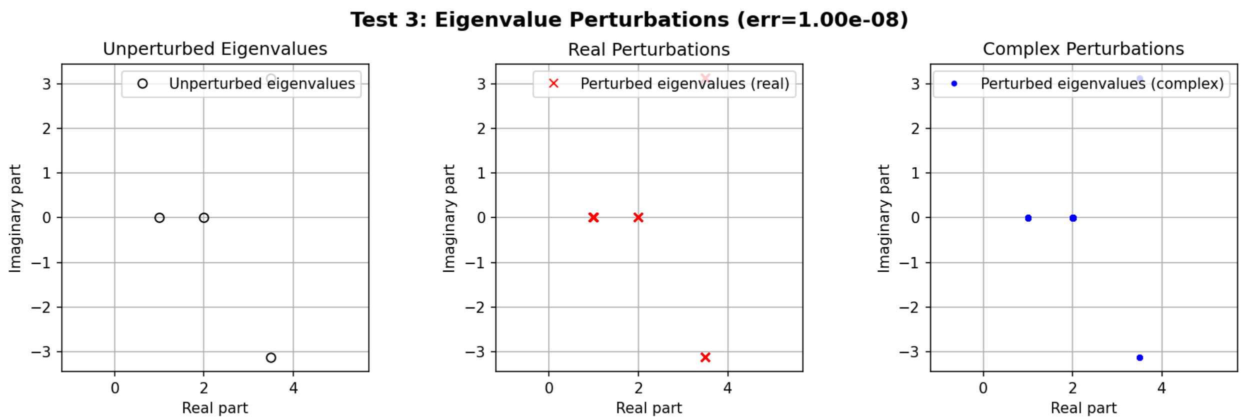
1. **Comparison with the bounds from §4.3.** In Test 2 all four eigenvalues of A are exactly $\lambda = 0$, and the condition numbers printed by the code are $\kappa_i = \infty$. This is the situation analyzed in §4.3 for multiple eigenvalues: a Jordan block has left and right eigenvectors e_n and e_1, so $1/|y^*x| = 1/|e_n^* e_1| = \infty$, and the first–order bound $|\delta\lambda| \leq \kappa_i \|\delta A\|_2$ becomes useless. Instead, Example 4.4 shows that for an $n \times n$ Jordan block a small perturbation of size ε can produce eigenvalues of magnitude about $\varepsilon^{1/n}$.:contentReference[oaicite:1]index=1 Here A is 4×4 with a quadruple eigenvalue at 0, and the complex perturbation plots indeed show the eigenvalues lying on a ring whose radius scales roughly like $\|\delta A\|^{1/4}$: when `err` $= 10^{-12}$, 10^{-10}, 10^{-8} the outer radius is about $2 \cdot 10^{-3}$, $6 \cdot 10^{-3}$ and $2 \cdot 10^{-2}$ respectively, consistent with a $\texttt{err}^{1/4}$ law rather than the linear behavior seen in Test 1. So Test 2 illustrates the *ill-conditioned, multiple-eigenvalue* regime of §4.3: very tiny perturbations can move the eigenvalues a comparatively large distance.

2. **Real versus complex perturbations.** With *complex* perturbations, the random δA explores all directions in $\mathbb{C}^{4\times 4}$, so the eigenvalue perturbations $\delta\lambda$ also explore all directions in the complex plane. This leads to the nearly circular annulus of eigenvalues surrounding the origin in the right–hand plots: for each fixed `err`, the pseudospectrum at level `err` is well approximated by a disk (or thin ring) of radius $\approx C\,\texttt{err}^{1/4}$. With *real* perturbations we restrict to real δA, so the characteristic polynomial has real coefficients and the four eigenvalues must appear as two complex conjugate pairs. This constraint, together with the specific Jordan structure of A, means that only certain directions in the complex plane are sampled, producing the striking "starburst" of rays in the middle plots rather than a filled disk. In both cases the radial size of the pseudospectrum is comparable, but the angular coverage is much less uniform for real perturbations.

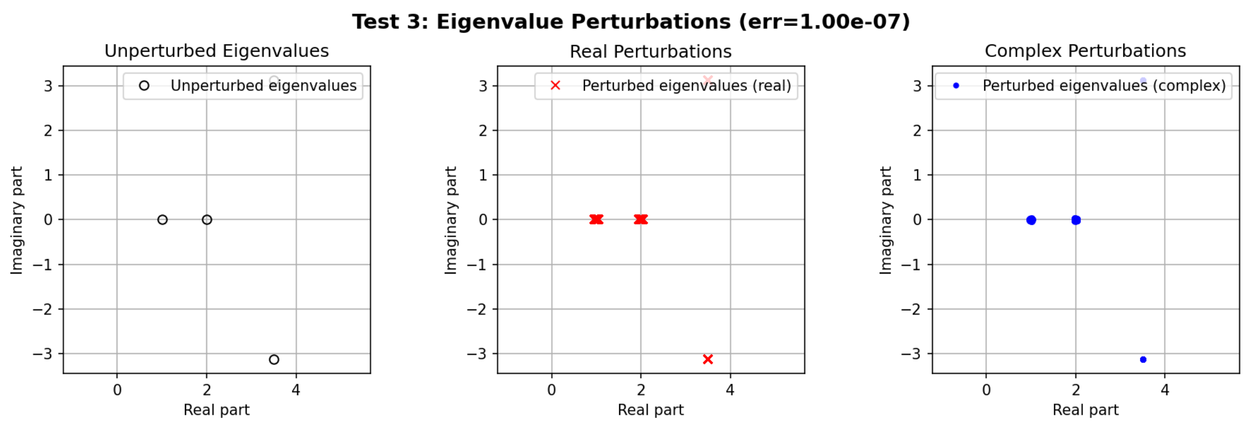
3. **Behavior as `err` $\to 0$.** As `err` decreases from 10^{-8} to 10^{-12} the radius of the pseudospectral ring shrinks, but much more slowly than linearly in `err`: the change is only

about one order of magnitude in radius for four orders of magnitude in `err`, consistent with the $\|\delta A\|^{1/4}$ behavior predicted for a 4×4 Jordan block. In the limit of mathematically exact arithmetic we would have $|\lambda| \sim C\,\texttt{err}^{1/4} \to 0$, so the ring would collapse to the origin, but extremely slowly compared with the well-conditioned case of Test 1. This slow collapse is precisely the manifestation of the infinite condition numbers of the multiple eigenvalue.

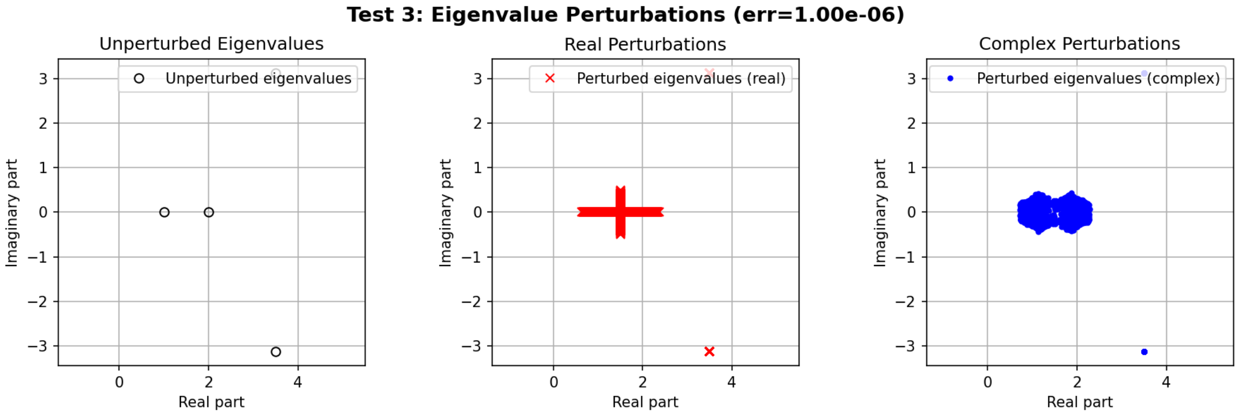
4. **Limiting size and number of correct digits.** For a double root one gets about half the digits, for a triple root about one third, and so on. Our matrix has a *quadruple* eigenvalue, so in double precision (about 16 decimal digits) we can expect only about $16/4 \approx 4$ correct digits in the eigenvalues. Equivalently, if the exact eigenvalue is 0 and the backward error in A is of order machine epsilon $u \approx 10^{-16}$, then the magnitude of the computed eigenvalues is typically of order $u^{1/4} \approx 10^{-4}$. Thus, even as $\texttt{err} \to 0$, the limiting size of the pseudospectral region is on the order of 10^{-4} in double precision, meaning that we can trust at most the first 3–4 digits in the statement "$\lambda = 0$"; finer distinctions (like whether $|\lambda|$ is 10^{-6} instead of 10^{-4}) are not numerically reliable for this highly ill-conditioned eigenvalue.



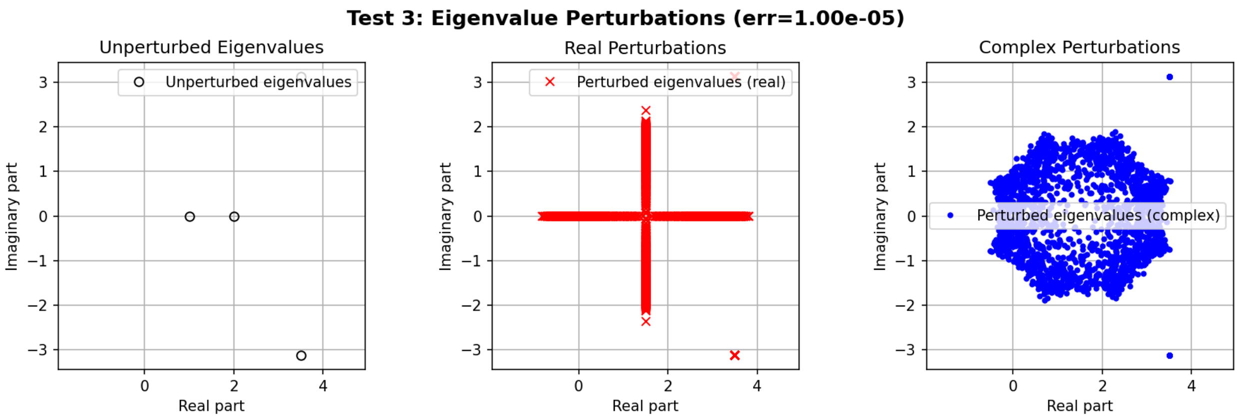
Eigenvalues	Condition Numbers
1.000000e+00+0.000000e+00j	1.000000e+06
2.000000e+00+0.000000e+00j	1.000000e+06
3.500000e+00+3.122499e+00j	1.212158e+02
3.500000e+00-3.122499e+00j	1.212158e+02



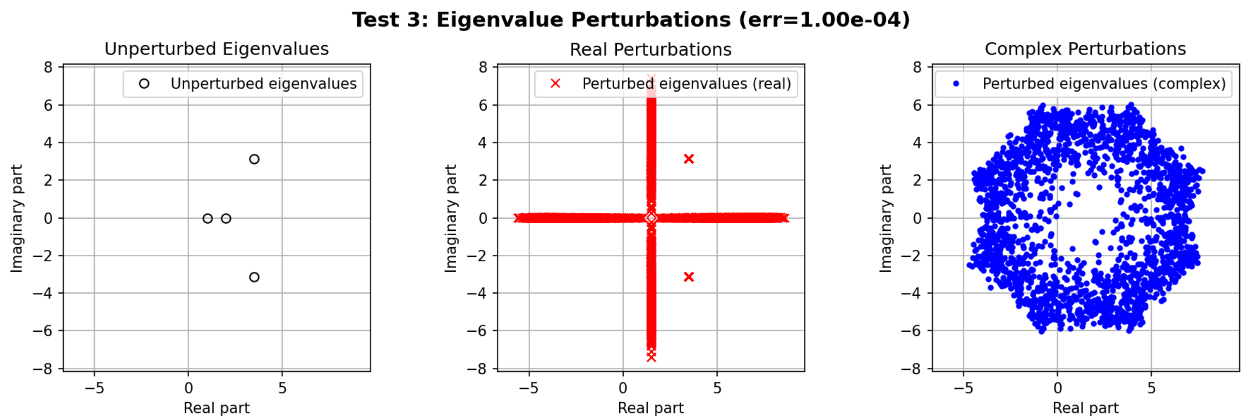
Eigenvalues	Condition Numbers
1.000000e+00+0.000000e+00j	1.000000e+06
2.000000e+00+0.000000e+00j	1.000000e+06
3.500000e+00+3.122499e+00j	1.212158e+02
3.500000e+00-3.122499e+00j	1.212158e+02



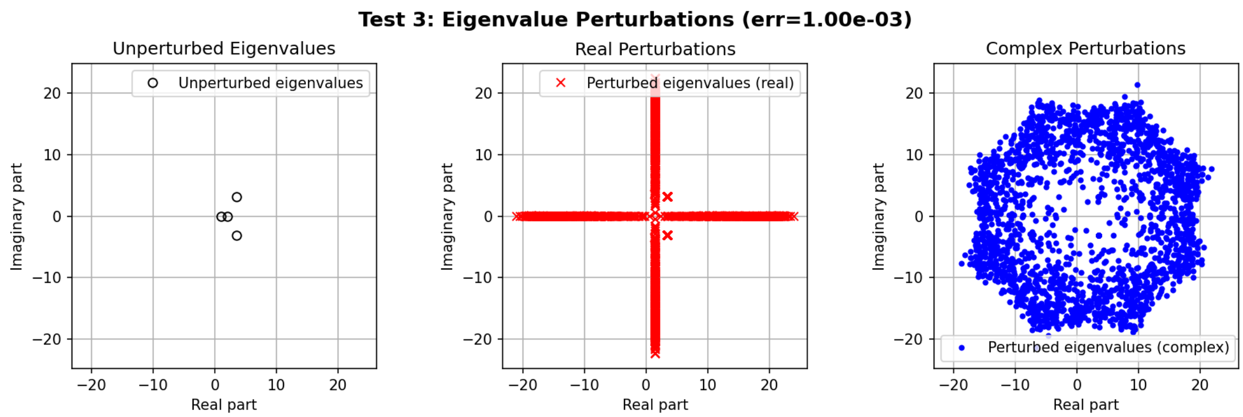
Eigenvalues	Condition Numbers
1.000000e+00+0.000000e+00j	1.000000e+06
2.000000e+00+0.000000e+00j	1.000000e+06
3.500000e+00+3.122499e+00j	1.212158e+02
3.500000e+00-3.122499e+00j	1.212158e+02



Eigenvalues	Condition Numbers
1.000000e+00+0.000000e+00j	1.000000e+06
2.000000e+00+0.000000e+00j	1.000000e+06
3.500000e+00+3.122499e+00j	1.212158e+02
3.500000e+00-3.122499e+00j	1.212158e+02



Eigenvalues	Condition Numbers
1.000000e+00+0.000000e+00j	1.000000e+06
2.000000e+00+0.000000e+00j	1.000000e+06
3.500000e+00+3.122499e+00j	1.212158e+02
3.500000e+00-3.122499e+00j	1.212158e+02



Eigenvalues	Condition Numbers
1.000000e+00+0.000000e+00j	1.000000e+06
2.000000e+00+0.000000e+00j	1.000000e+06
3.500000e+00+3.122499e+00j	1.212158e+02
3.500000e+00-3.122499e+00j	1.212158e+02

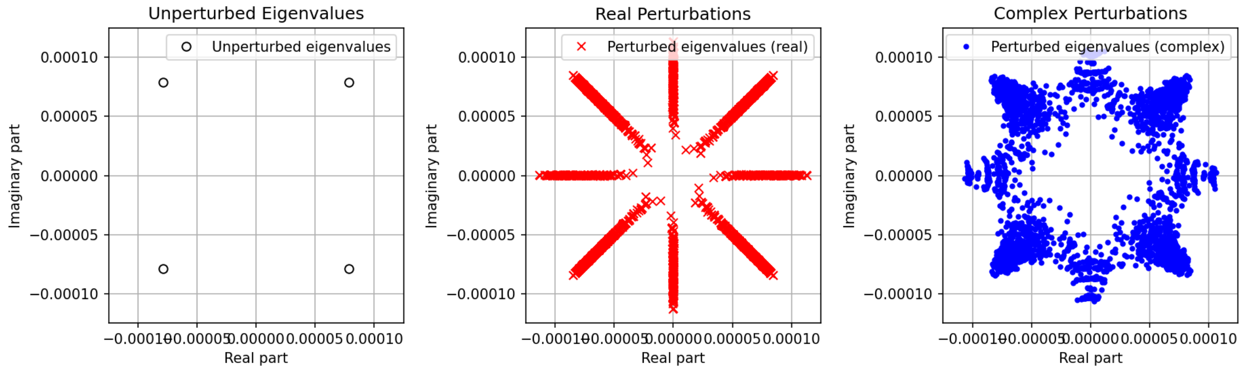
1. **Comparison with the perturbation bounds.** In Test 3 the unperturbed eigenvalues are $\lambda_1 = 1$, $\lambda_2 = 2$, and $\lambda_{3,4} = 3.5 \pm 3.12\,i$. The condition numbers printed under the plots show that λ_1 and λ_2 are extremely ill conditioned ($\kappa_1 = \kappa_2 \approx 10^6$), while the complex conjugate pair is moderately conditioned ($\kappa_3 = \kappa_4 \approx 1.2 \times 10^2$). The usual first-order perturbation theory says that eigenvalues of $A + \Delta A$ should lie in discs of radius about $\kappa_i \|\Delta A\| \sim \kappa_i\,\texttt{err}$ centered at the unperturbed eigenvalues. This is exactly what the plots show. For the complex pair, the clouds of eigenvalues under complex perturbations remain small and roughly circular even when $\texttt{err} = 10^{-3}$, because κ_3, κ_4 are only $\mathcal{O}(10^2)$. In contrast, the pseudospectral region associated with the real eigenvalues at 1 and 2 is enormous: even for relatively small $\texttt{err} = 10^{-4}$ or 10^{-5} the perturbed eigenvalues spread far from 1 and 2, and for $\texttt{err} = 10^{-3}$ they wander over a large portion of the plotting window. This behavior is consistent with the factor $\kappa_1 = \kappa_2 \approx 10^6$ in the bound $|\delta\lambda| \lesssim \kappa_i\,\texttt{err}$.

2. **Real versus complex perturbations.** For real perturbations the perturbation matrices are restricted to be real, so the characteristic polynomial has real coefficients and the complex eigenvalues still come in conjugate pairs. This restriction forces the eigenvalue motion into certain preferred directions in the complex plane. In the plots this shows up as the striking "cross" pattern: the most sensitive eigenvalues near 1 and 2 are pushed far along the real axis or far along the imaginary axis, giving long horizontal and vertical streaks rather than a filled disc. With complex perturbations there is no such restriction, so the perturbations are essentially isotropic; the corresponding eigenvalues fill out more circular, blob-shaped regions that give a better picture of the full pseudospectrum. The moderately conditioned pair near $3.5 \pm 3.12i$ already looks nearly circular in both cases; the dramatic difference is for the highly ill-conditioned real eigenvalues.

3. **Behavior as `err` goes to zero.** Comparing the plots from $\texttt{err} = 10^{-3}$ down to $\texttt{err} = 10^{-8}$, the regions occupied by the eigenvalues shrink as `err` decreases. For the complex pair, the diameter of the cloud is roughly proportional to `err`: when `err` is reduced by a factor of 10, the spread of those eigenvalues shrinks by roughly the same factor, and for very small `err` the cluster around $3.5 \pm 3.12i$ is barely visible at the scale of the plots. The real eigenvalues at 1 and 2 also move less as `err` decreases, but because their condition numbers are so large they remain much more spread out than the complex pair for the same `err`: even at $\texttt{err} = 10^{-7}$ the clouds around 1 and 2 are still noticeably larger than those around $3.5 \pm 3.12i$. In the idealized limit $\texttt{err} \to 0$ (with exact arithmetic) all of these regions would collapse to the exact eigenvalues, but the rate of collapse is governed by the corresponding condition numbers.

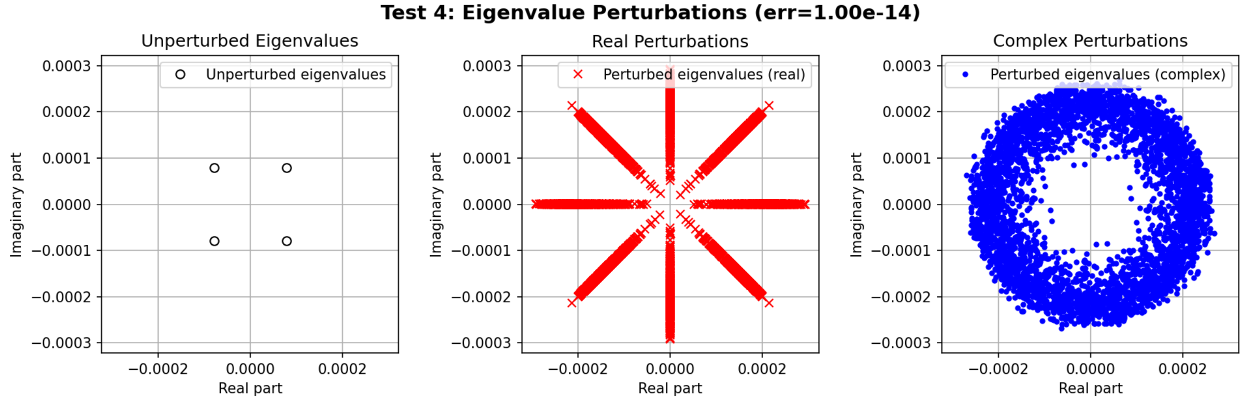
4. **Limiting size and number of correct digits.** In floating-point arithmetic a backward-stable eigenvalue algorithm effectively perturbs the matrix by an amount on the order of $u\|A\|$, where u is the unit roundoff ($u \approx 10^{-16}$ in double precision). Substituting $\|\Delta A\| \sim u$ into the bound $|\delta\lambda| \lesssim \kappa_i \|\Delta A\|$ suggests that the limiting eigenvalue error is about $\kappa_i u$. For the two highly ill-conditioned eigenvalues with $\kappa \approx 10^6$, this gives a limiting absolute error of order $10^6 \times 10^{-16} = 10^{-10}$, so we can trust only about 6 correct decimal digits in those eigenvalues. For the complex conjugate pair with $\kappa \approx 1.2 \times 10^2$, the limiting error is about $10^2 \times 10^{-16} \approx 10^{-14}$, so roughly 14 digits are reliable. Thus Test 3 illustrates a mixed situation: two eigenvalues are so ill conditioned that only a handful of digits are meaningful, whereas the other two eigenvalues are accurate to essentially full double-precision accuracy.

Test 4: Eigenvalue Perturbations (err=1.00e-16)



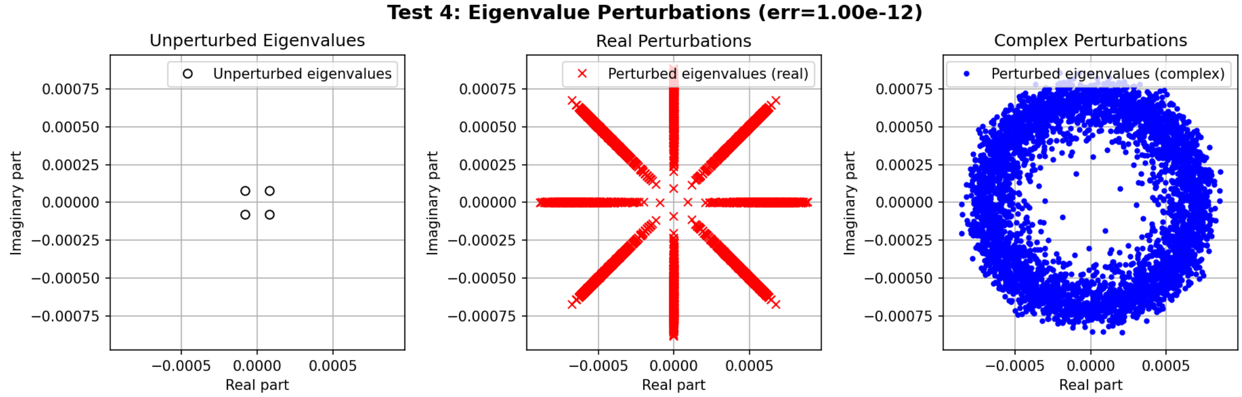
Eigenvalues	Condition Numbers
-7.878899e-05+7.863573e-05j	1.812296e+11
-7.864667e-05-7.877820e-05j	1.812293e+11
7.878914e-05-7.865745e-05j	1.811793e+11
7.864651e-05+7.879992e-05j	1.811796e+11

Test 4: Eigenvalue Perturbations (err=1.00e-14)

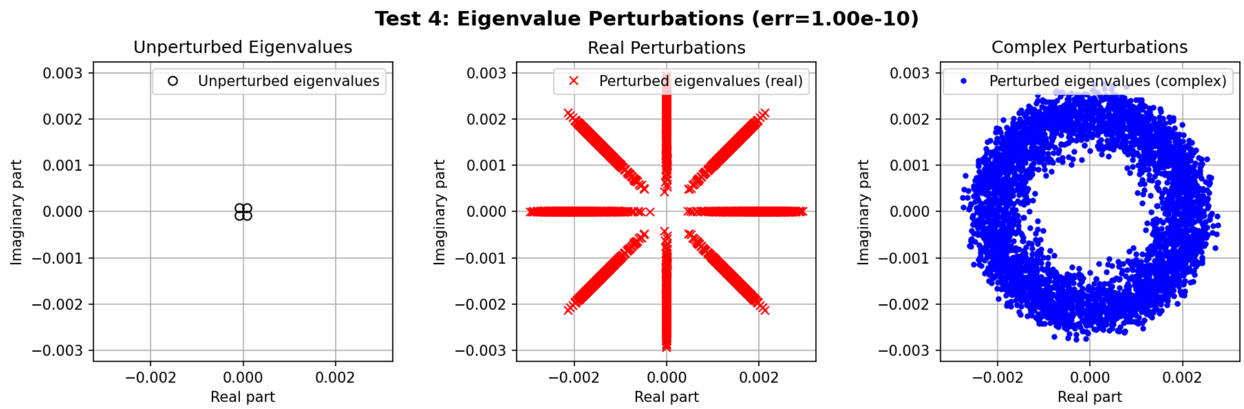


Eigenvalues	Condition Numbers
-7.878899e-05+7.863573e-05j	1.812296e+11
-7.864667e-05-7.877820e-05j	1.812293e+11
7.878914e-05-7.865745e-05j	1.811793e+11
7.864651e-05+7.879992e-05j	1.811796e+11

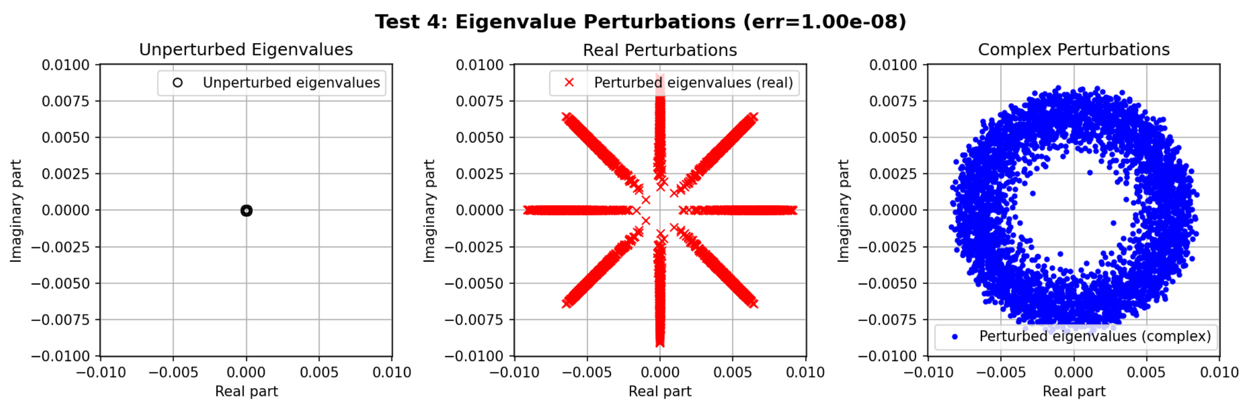
Test 4: Eigenvalue Perturbations (err=1.00e-12)



Eigenvalues	Condition Numbers
-7.878899e-05+7.863573e-05j	1.812296e+11
-7.864667e-05-7.877820e-05j	1.812293e+11
7.878914e-05-7.865745e-05j	1.811793e+11
7.864651e-05+7.879992e-05j	1.811796e+11



Eigenvalues	Condition Numbers
-7.878899e-05+7.863573e-05j	1.812296e+11
-7.864667e-05-7.877820e-05j	1.812293e+11
7.878914e-05-7.865745e-05j	1.811793e+11
7.864651e-05+7.879992e-05j	1.811796e+11



Eigenvalues	Condition Numbers
-7.878899e-05+7.863573e-05j	1.812296e+11
-7.864667e-05-7.877820e-05j	1.812293e+11
7.878914e-05-7.865745e-05j	1.811793e+11
7.864651e-05+7.879992e-05j	1.811796e+11

1. **Comparison with the bounds from Section 4.3.** In Test 4 there are four simple eigenvalues, all very close to each other and to the origin:

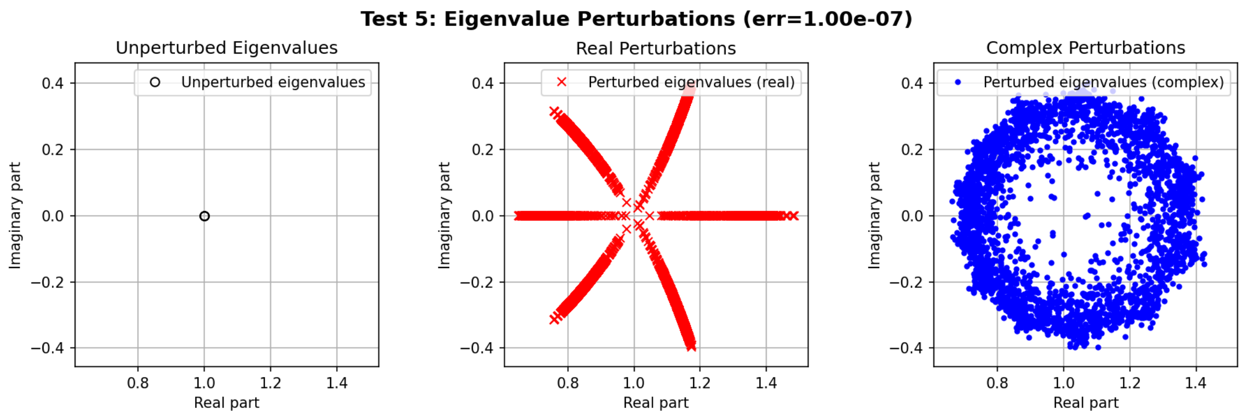
$$\lambda_1, \ldots, \lambda_4 \approx 7.9 \times 10^{-5} \pm 7.9 \times 10^{-5} i,$$

and their eigenvalue condition numbers are all about $\kappa_i \approx 1.8 \times 10^{11}$. The bounds in Section 4.3 say that, for a perturbation ΔA with norm $\|\Delta A\| \approx \texttt{err}$, each eigenvalue of $A + \Delta A$ should lie in a disc $|\lambda - \lambda_i| \lesssim \kappa_i \, \texttt{err}$. Because κ_i is enormous, even tiny values of `err` give discs much larger than the separation between the four eigenvalues. This is exactly what the complex–perturbation plots show: instead of four small, well-separated clusters, the eigenvalues of the perturbed matrices fill out a single large annulus surrounding the origin. As `err` increases from 10^{-16} to 10^{-8}, the radius of this annulus increases by many orders of magnitude and quickly becomes much larger than the distance of the unperturbed eigenvalues from the origin, which is consistent with the factor κ_i in the perturbation bound.

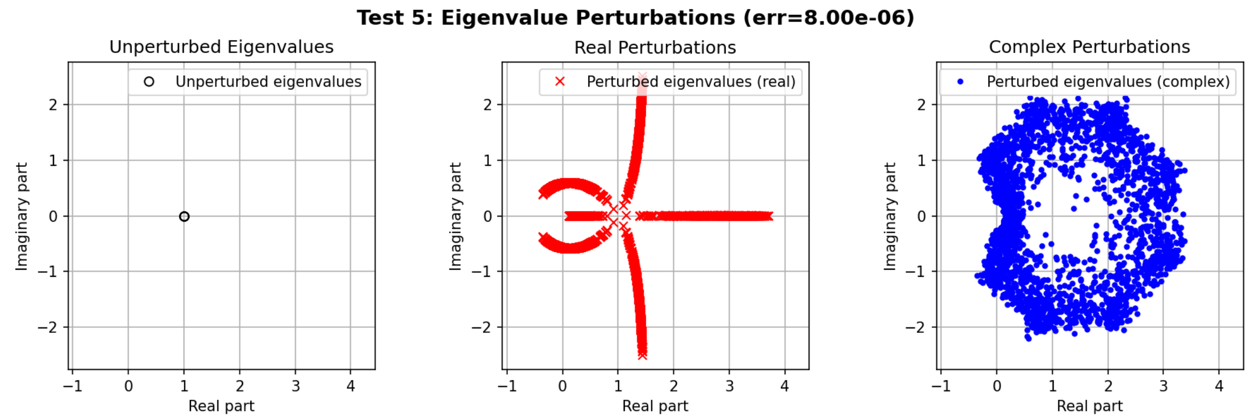
2. **Difference between real and complex perturbations.** For *complex* perturbations, the random matrices ΔA explore all directions in the complex space of matrices, so the eigenvalue changes sample all directions in the complex plane. Consequently, the perturbed eigenvalues form nearly circular rings around the origin, giving a good picture of the full pseudospectrum at radius $\|\Delta A\| \approx$ `err`. For *real* perturbations, ΔA is constrained to be real. The characteristic polynomial then has real coefficients and the four eigenvalues must appear as two complex conjugate pairs. This restriction forces the eigenvalue motion into preferred directions, producing the "starburst" pattern in the real–perturbation plots: the points lie along eight rays rather than filling out a solid disc. The radial extent of the cloud is comparable in the two cases, but the angular distribution is much less uniform for real perturbations.

3. **Behavior as `err` goes to zero.** As `err` decreases from 10^{-8} to 10^{-16}, the radius of the pseudospectral annulus shrinks toward the cluster of unperturbed eigenvalues. For large `err` the annulus is so wide that the perturbed eigenvalues are spread over a region whose size is orders of magnitude larger than the original eigenvalues. As `err` is reduced by factors of 10^2, the diameter of the annulus contracts correspondingly; by the time `err` reaches 10^{-16} the cloud has radius of order 10^{-5}–10^{-4}, comparable to the magnitude of the eigenvalues themselves. In exact arithmetic, letting `err` $\to 0$ would cause the annulus to collapse to four points at the unperturbed eigenvalues, but because the eigenvalues are so ill conditioned this collapse is extremely slow.

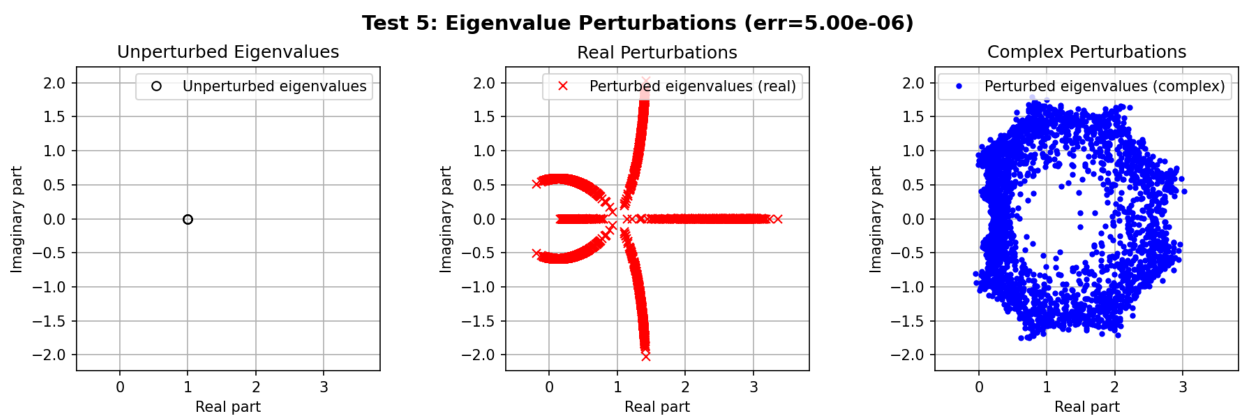
4. **Limiting size and number of correct digits.** In floating–point arithmetic a backward–stable algorithm effectively introduces a perturbation $\|\Delta A\|$ of order $u\|A\|$, where $u \approx 10^{-16}$ is the unit roundoff. The perturbation bound then predicts an eigenvalue error of size $|\delta\lambda| \approx \kappa_i u\|A\|$. With $\kappa_i \approx 1.8 \times 10^{11}$ and $\|A\|$ of order one, this gives $|\delta\lambda| \sim 10^{-5}$. Since the eigenvalues themselves are only about 10^{-4} in magnitude, a perturbation of size 10^{-5} corresponds to a relative error of roughly ten percent: the eigenvalues can move by an amount comparable to one significant digit in their value. Thus, even as `err` is taken as small as machine precision, the limiting pseudospectral region has radius of order 10^{-5}, and only about one significant digit of each of these eigenvalues is truly reliable; their exact locations beyond this coarse scale are numerically meaningless.



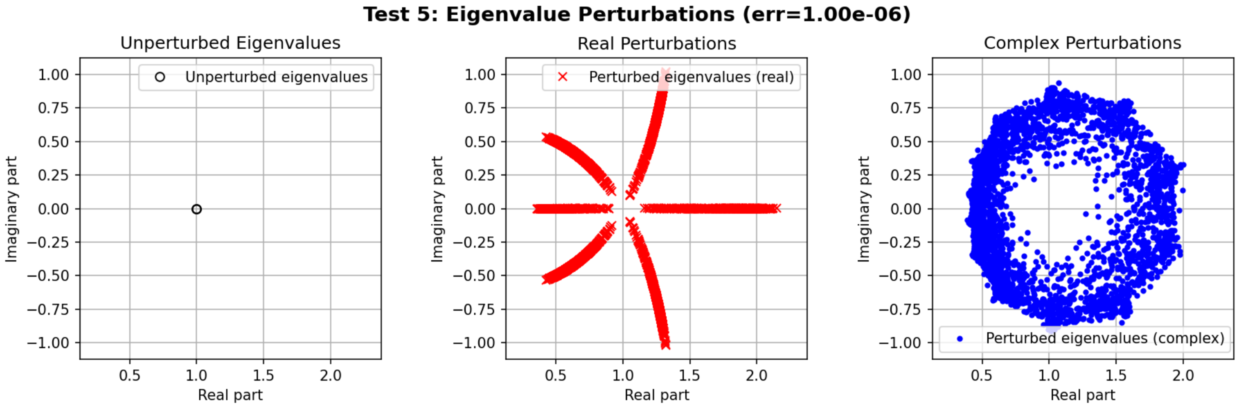
Eigenvalues	Condition Numbers
1.000000e+00+0.000000e+00j	9.444734e+21
1.000000e+00+0.000000e+00j	2.028241e+37
1.000000e+00+0.000000e+00j	2.028241e+37



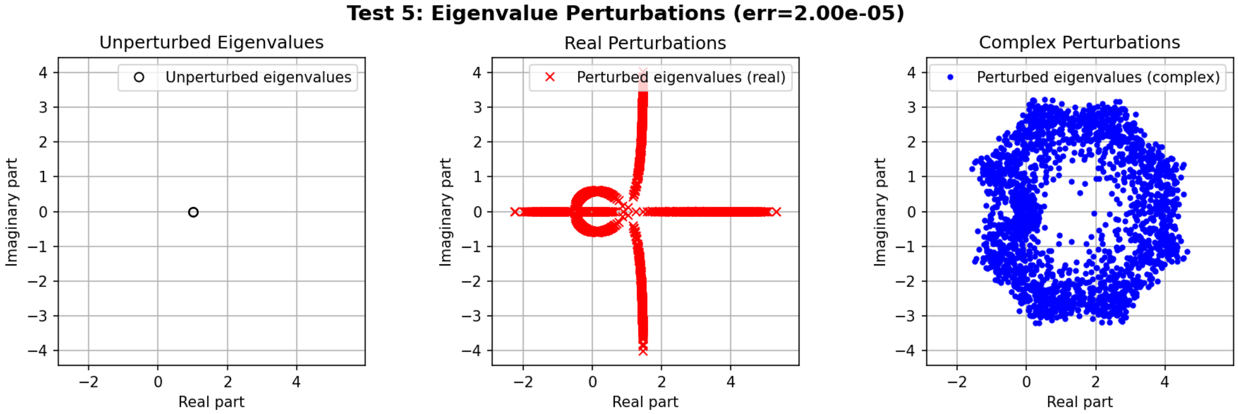
Eigenvalues	Condition Numbers
1.000000e+00+0.000000e+00j	9.444734e+21
1.000000e+00+0.000000e+00j	2.028241e+37
1.000000e+00+0.000000e+00j	2.028241e+37



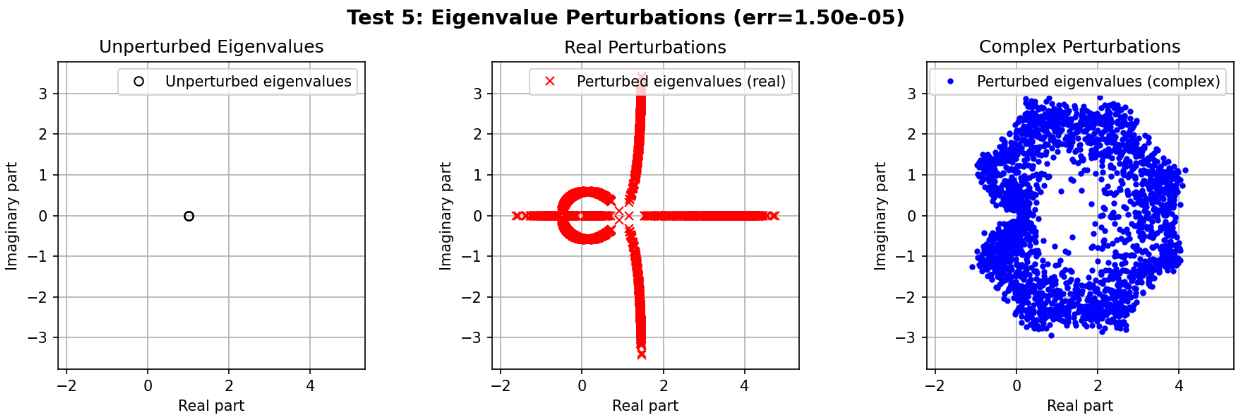
Eigenvalues	Condition Numbers
1.000000e+00+0.000000e+00j	9.444734e+21
1.000000e+00+0.000000e+00j	2.028241e+37
1.000000e+00+0.000000e+00j	2.028241e+37



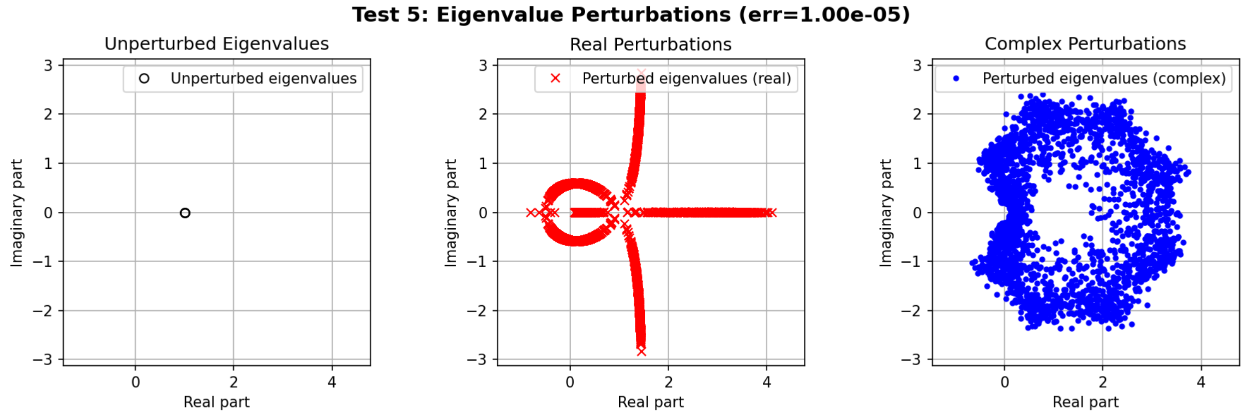
Eigenvalues	Condition Numbers
1.000000e+00+0.000000e+00j	9.444734e+21
1.000000e+00+0.000000e+00j	2.028241e+37
1.000000e+00+0.000000e+00j	2.028241e+37



Eigenvalues	Condition Numbers
1.000000e+00+0.000000e+00j	9.444734e+21
1.000000e+00+0.000000e+00j	2.028241e+37
1.000000e+00+0.000000e+00j	2.028241e+37



Eigenvalues	Condition Numbers
1.000000e+00+0.000000e+00j	9.444734e+21
1.000000e+00+0.000000e+00j	2.028241e+37
1.000000e+00+0.000000e+00j	2.028241e+37



Eigenvalues	Condition Numbers
1.000000e+00+0.000000e+00j	9.444734e+21
1.000000e+00+0.000000e+00j	2.028241e+37
1.000000e+00+0.000000e+00j	2.028241e+37

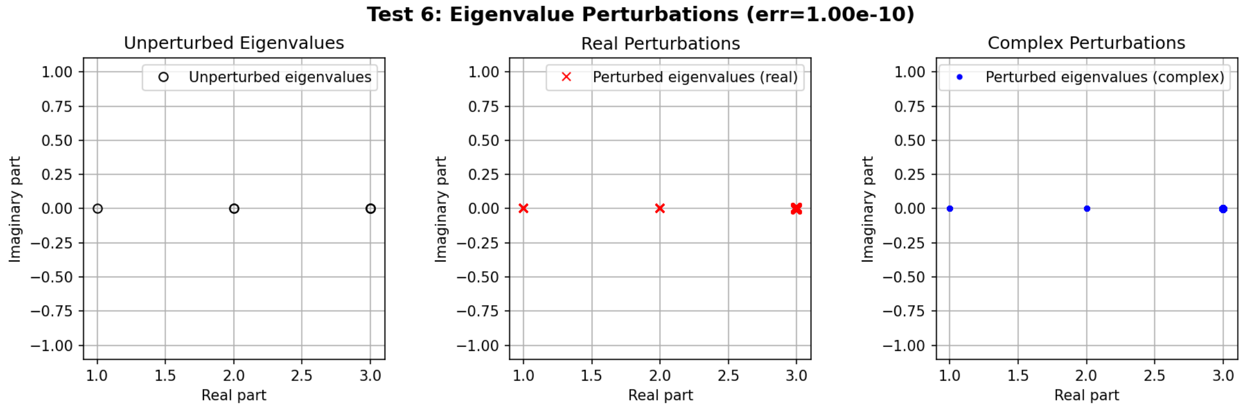
1. **Comparison with the bounds from Section 4.3.** In Test 5 the matrix has a triple eigenvalue at $\lambda = 1$: the eigenvalue table lists three identical eigenvalues, and the associated condition numbers are enormous ($\kappa \approx 10^{22}, 10^{37}, 10^{37}$). The first–order bound from Section 4.3 says that eigenvalues of $A+\Delta A$ lie in discs $|\lambda - \lambda_i| \lesssim \kappa_i \|\Delta A\| \approx \kappa_i\, \texttt{err}$. With κ_i this large, these discs would have radius far exceeding 1 even for very small `err`, so the linear bound is essentially useless: it predicts that the eigenvalues can move almost anywhere in the complex plane. The plots show the more refined reality for a multiple eigenvalue: for complex perturbations the eigenvalues of the perturbed matrices lie on a roughly circular annulus centered at 1, whose radius grows much more slowly than linearly with `err`. As `err` varies from 10^{-7} up to 2×10^{-5}, the annulus radius changes only by about one order of magnitude, consistent with a scaling of the form $|\delta\lambda| \sim C\, \texttt{err}^{1/3}$ that is typical for a triple eigenvalue (rather than $|\delta\lambda| \sim C\, \texttt{err}$ as in the simple, well–conditioned case).

2. **Real versus complex perturbations.** For complex perturbations, the random matrices ΔA explore all directions in the space of complex matrices. The resulting eigenvalue changes also explore all directions in the complex plane, and the perturbed eigenvalues fill out a fairly uniform ring around $\lambda = 1$. This ring is a good picture of the pseudospectrum at level $\|\Delta A\| \approx \texttt{err}$. For real perturbations, ΔA is constrained to be real, so the characteristic polynomial has real coefficients and the three eigenvalues are either all real or one real plus a complex conjugate pair. This restriction forces the eigenvalue motion into special curves rather than a full ring. In the real–perturbation plots one sees a "pinwheel" pattern: branches of eigenvalues spiralling away from 1 along certain preferred directions, together with a smaller loop close to the unperturbed value. The radial scale of this pattern is comparable to that of the complex–perturbation annulus, but the angular coverage is much less uniform.

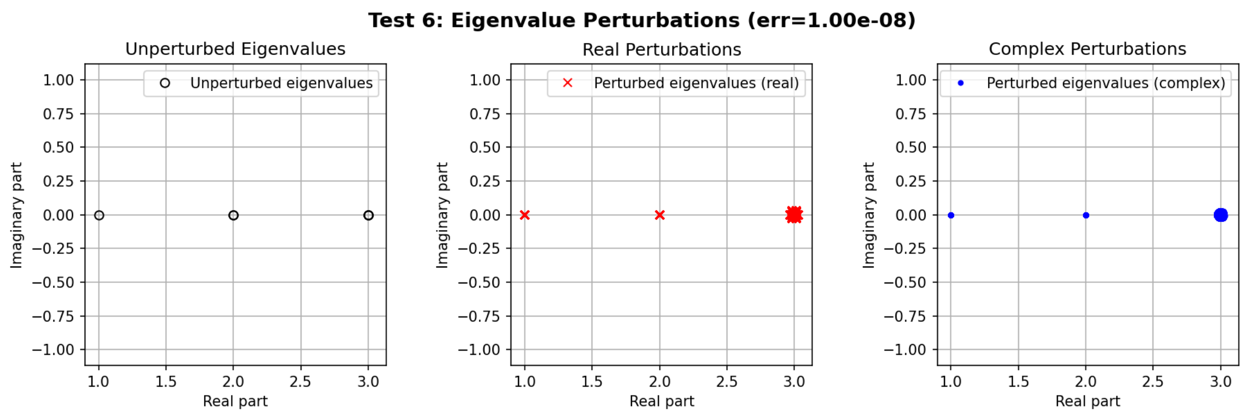
3. **Behavior as `err` goes to zero.** As `err` decreases (from 2×10^{-5} down to 10^{-7} and 10^{-6}), both the annulus in the complex case and the pinwheel in the real case contract toward the unperturbed eigenvalue at 1. However, the contraction is markedly

slower than linear: reducing `err` by more than an order of magnitude produces only a modest shrinkage of the pseudospectral region. This slow collapse reflects the multiple–eigenvalue behavior: eigenvalue displacement is governed by a fractional power of `err`, so even very small perturbations can move the eigenvalues a macroscopic distance. In the limit of exact arithmetic and `err` $\to 0$ the regions would of course collapse to the point $\lambda = 1$, but the pictures make clear that this happens very slowly.

4. **Limiting size and number of correct digits.** In floating–point arithmetic a backward–stable algorithm effectively introduces a matrix perturbation of size $\|\Delta A\| \sim u\|A\|$, where u is the unit roundoff (about 10^{-16} in double precision). For a triple eigenvalue, the generic perturbation theory predicts eigenvalue errors of order $u^{1/3}$. Numerically this is about $10^{-16/3} \approx 10^{-5}$. Since the exact eigenvalue is $\lambda = 1$, this means that the computed eigenvalues will typically be scattered in a disc of radius about 10^{-5} around 1, even if the algorithm is perfectly stable. Thus only roughly 5 decimal digits of $\lambda = 1$ can be trusted; finer digits are overwhelmed by the sensitivity of the multiple eigenvalue to tiny perturbations. The annulus seen in the smallest-`err` plots is therefore close to its limiting size, set not by the user–chosen `err` but by the intrinsic rounding error of the arithmetic.



Eigenvalues	Condition Numbers
1.000000e+00+0.000000e+00j	1.000000e+00
2.000000e+00+0.000000e+00j	2.251800e+15
2.000000e+00+0.000000e+00j	2.251800e+15
3.000000e+00+0.000000e+00j	1.844735e+19
3.000000e+00+0.000000e+00j	2.253601e+34
3.000000e+00+0.000000e+00j	2.253601e+34



Eigenvalues	Condition Numbers
1.000000e+00+0.000000e+00j	1.000000e+00
2.000000e+00+0.000000e+00j	2.251800e+15
2.000000e+00+0.000000e+00j	2.251800e+15
3.000000e+00+0.000000e+00j	1.844735e+19
3.000000e+00+0.000000e+00j	2.253601e+34
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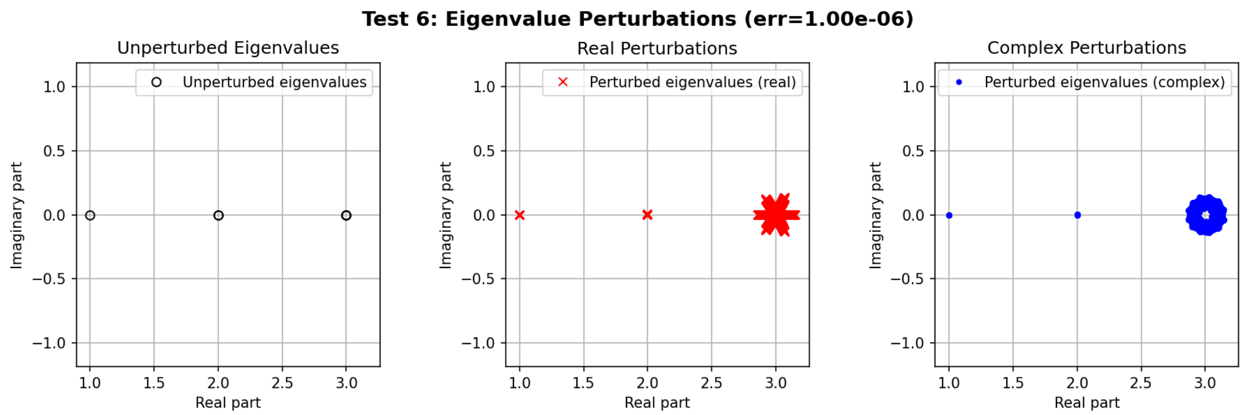

Eigenvalues	Condition Numbers
1.000000e+00+0.000000e+00j	1.000000e+00
2.000000e+00+0.000000e+00j	2.251800e+15
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3.000000e+00+0.000000e+00j	1.844735e+19
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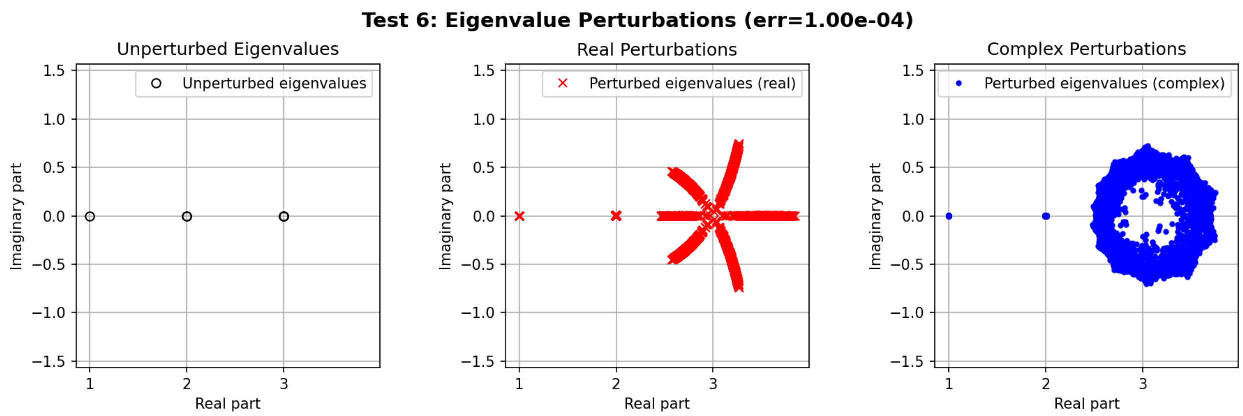

Eigenvalues	Condition Numbers
1.000000e+00+0.000000e+00j	1.000000e+00
2.000000e+00+0.000000e+00j	2.251800e+15
2.000000e+00+0.000000e+00j	2.251800e+15
3.000000e+00+0.000000e+00j	1.844735e+19
3.000000e+00+0.000000e+00j	2.253601e+34
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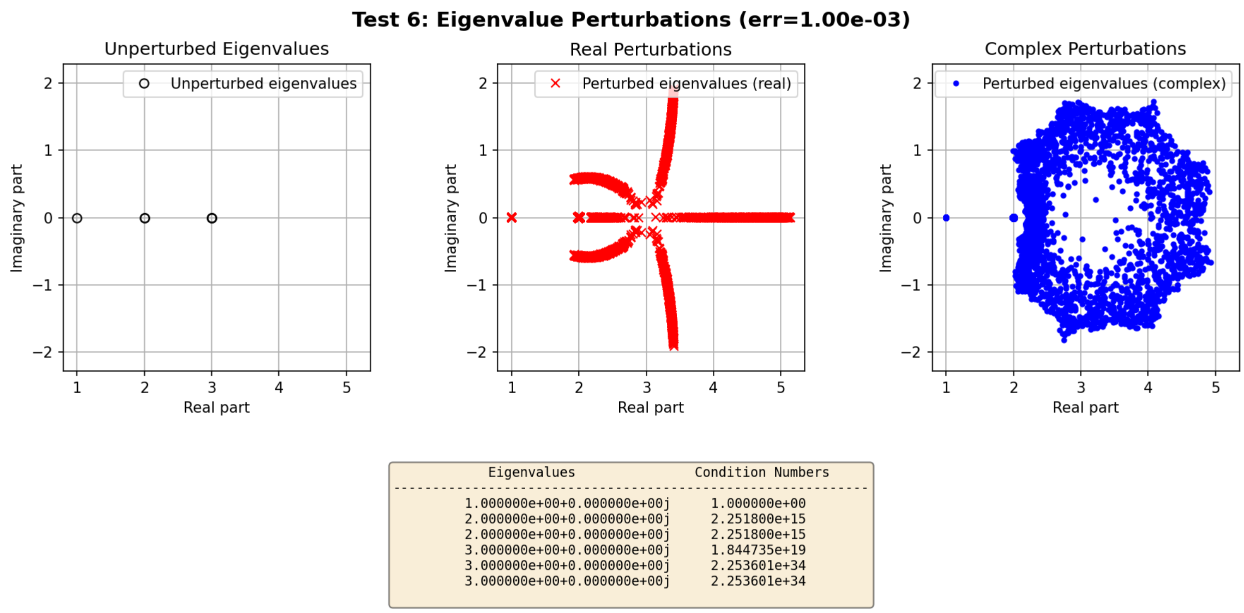

1. **Comparison with the perturbation bounds from Section 4.3.** In Test 6 the unperturbed eigenvalues are

$$\lambda_1 = 1, \quad \lambda_2 = \lambda_3 = 2, \quad \lambda_4 = \lambda_5 = \lambda_6 = 3,$$

with condition numbers $\kappa_1 \approx 10^5$, $\kappa_2 = \kappa_3 \approx 2.3 \times 10^{15}$, $\kappa_4 \approx 1.8 \times 10^{19}$, $\kappa_5 = \kappa_6 \approx 2.3 \times 10^{34}$. The standard bound in Section 4.3 says that each eigenvalue of $A + \Delta A$ lies in a disc $|\lambda - \lambda_i| \lesssim \kappa_i \|\Delta A\|$, so for the simple eigenvalue at 1 this predicts a modest disc of radius about 10^5`err`, while for the highly ill-conditioned eigenvalues near 2 and especially 3 it predicts astronomically large discs and is essentially useless. The complex–perturbation plots reflect this: the eigenvalue at 1 hardly moves at all, while the cluster near 2 spreads noticeably and the cluster near 3 occupies a very large annulus whose radius grows much more slowly than linearly in `err`, as expected for multiple eigenvalues (roughly like $\texttt{err}^{1/2}$ for the double eigenvalue at 2 and $\texttt{err}^{1/3}$ for the triple eigenvalue at 3). Thus Test 6 combines a reasonably well-conditioned simple eigenvalue with extremely ill-conditioned multiple eigenvalues, and the pseudospectrum is dominated by the latter.

2. **Difference between real and complex perturbations.** For complex perturbations, the random matrices ΔA explore all directions in the complex matrix space, so the perturbed eigenvalues fill out approximately circular annuli around the clusters at 2 and 3, while the eigenvalue at 1 remains a tiny dot on the real axis. In contrast, for real perturbations the matrix ΔA is real and the characteristic polynomial has real coefficients, so complex eigenvalues must occur in conjugate pairs and the motion of the eigenvalues is constrained to special curves. This yields the characteristic "pinwheel" patterns in the real–perturbation plots: the multiple eigenvalues near 3 move along several spiralling rays rather than filling out a full ring. The radial scale of the motion is comparable in the real and complex cases (set by the sensitivity of the multiple eigenvalues), but the angular distribution is much less uniform for real perturbations.

3. **Behavior as `err` goes to zero.** As `err` decreases from 10^{-3} to 10^{-10}, the pseudospectral regions shrink. For the simple eigenvalue at 1 the shrinking is essentially linear: even at `err`$= 10^{-3}$ the motion is barely visible on the plots, and by `err`$= 10^{-6}$ or smaller it is negligible. The double eigenvalue at 2 shrinks more slowly; its perturbed values form a small line segment or short arc for moderate `err`, collapsing toward the real axis as `err` is reduced. The triple eigenvalue at 3 shrinks slowest of all: for `err`$= 10^{-3}$ the complex–perturbation eigenvalues form a large annulus around 3, and only when `err` reaches 10^{-8}–10^{-10} does this annulus collapse into a tight cluster near the unperturbed value. This hierarchy of shrinking rates matches the theory that multiple eigenvalues respond to perturbations with a fractional power of $\|\Delta A\|$, while simple eigenvalues respond linearly.

4. **Limiting size of the regions and number of correct digits.** In double precision, a backward-stable eigenvalue algorithm perturbs the matrix by about $u\|A\|$ with $u \approx 10^{-16}$. Combining this with the condition numbers suggests that the effective eigenvalue errors at `err` below machine precision are roughly

$$|\delta\lambda_1| \sim \kappa_1 u \approx 10^{-11}, \quad |\delta\lambda_{2,3}| \sim \kappa_2 u \approx 10^{-1}, \quad |\delta\lambda_{4,5,6}| \sim \kappa_5 u \gg 1.$$

   Thus the eigenvalue at 1 is accurate to about $-\log_{10}(\kappa_1 u) \approx 11$ decimal digits. The double eigenvalue at 2 has $\kappa \approx 10^{15}$, so κu is of order 10^{-1}; at best one decimal digit is reliable, and even that may be optimistic. For the triple eigenvalue at 3 the condition numbers are so large ($\kappa \approx 10^{19}$ to 10^{34}) that κu greatly exceeds 1; in practice no significant digits of these eigenvalues are trustworthy. Consequently, as `err`$\to 0$ the pseudospectral region around $\lambda = 1$ eventually shrinks to a tiny disc of radius about 10^{-11}, while the regions around 2 and 3 flatten out at much larger radii set by rounding error, meaning those eigenvalues are almost completely unreliable numerically.

The source code for all experiments is on Github.